Application of the Random Ray Method to Variance Reduction in Radiation Shielding Problems

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Abstract

Deep penetration radiation shielding problems often use Monte Carlo augmented with weight windows to obtain sufficiently low variances in acceptable run times. In this paper, the random ray method was proposed as an approach to generating weight windows. Compared to deterministic alternatives, it allows for straightforward spatial discretisation and continuously samples angular phase space. The methods implemented were tested on two shielding problems, examining global variance reduction and source-detector variance reduction. In both cases, the figure of merit of the random ray based methods is at least 50 times higher than analog Monte Carlo, and in the Global Variance Reduction model it is comparable to that of MAGIC, a popular fully Monte Carlo approach to weight window generation. The methods implemented are also robust to modelling choices like the source of multi group cross sections (NJOY or Monte Carlo), scattering anisotropy and the order of the random ray spatial source representation.

Keywords — Variance Reduction, The Random Ray Method, FW-CADIS, CADIS

I. INTRODUCTION

Some applications of interest for nuclear engineering, such as fusion reactor design and radiation shielding problems, are generally simulated with Monte Carlo (MC) methods and require variance reduction (VR) to produce acceptable results. VR techniques normally work by producing and reading importance maps or weight windows to direct particles towards the locations of interest. One method to employ weight window maps is in combination with particle splitting and rouletting: splitting particles when their weight is higher than the local upper weight window bound, and applying Russian roulette when their weight is less than the lower weight window bound. Alternatively, weight windows can inform techniques like exponential transforms [1] or adaptive multilevel splitting [2]. However, the generation of appropriate weight windows is nontrivial, and there are several possible ways to do it.

The most common way to produce weight window maps is Consistent Adjoint-Driven Importance Sampling (CADIS), which relates the weight boundaries in each phase space region to a particle's local importance, i.e, the adjoint flux [3]. The adjoint flux is normally obtained with a deterministic solver. For example, MCNP [4] is coupled with the ad hoc software ADVANTG [5], which produces importance maps by running the discrete ordinates (S_N) code Denovo [6]. Both TRIPOLI [7] and MCBEND [8] have an in-built adjoint diffusion solver. Clearly, all these codes need to have multi group cross sections available for the problem of interest before running their deterministic solvers.

Serpent [9], on the other hand, uses an internal adjoint response matrix solver to generate importance maps [10] which is CADIS-inspired but employs different definitions. The importance in this case is related to the adjoint current entering each cell of the importance map. This solver does not require multi group cross sections, but coupling coefficients between different mesh cells, tallied by Monte Carlo. Since the response matrix mesh must cover all the regions of interest, and particles might not reach each cell of the mesh during an analog Monte Carlo calculation, as is required for tallying the coupling coefficients, this approach requires an iterative solution. Each iteration, the weight windows calculated by the response matrix solver are used in a Monte Carlo run to calculate the new coefficients, until the particle population covers the whole geometry and the final importance map

can be produced.

When the goal of a variance reduction calculation is to minimise the variance at a specific location, the adjoint source for the deterministic calculation is simply the detector cross section at the detector's position in phase space. If the calculation aims at obtaining a uniform statistical uncertainty throughout the problem geometry (Global Variance Reduction or GVR), or a low uncertainty at multiple detectors, Forward-Weighted CADIS (FW-CADIS) [11] is used. FW-CADIS requires two steps to produce weight windows: first, a forward transport calculation is run to obtain the forward flux, used to generate an adjoint source; then, an adjoint calculation generates an adjoint flux solution, used to construct importance maps.

In the case of GVR, alternative methods not based on an adjoint flux solution were also proposed in the past. For example, in [12], it is shown that if the goal is to obtain a uniform density of simulated particles throughout the model phase space, the weight distribution w must simply be proportional to the physical neutron density. One criticism to this approach is that the distribution of the population might not reflect the distribution of the importance, i.e., the contributions to the desired tally. However, this approach was shown to produce good results in [12] and [13], where the forward flux was used as a proxy for the neutron density. The forward flux solution can be generated in multiple ways: [12] uses a quasi-diffusion solution, where the necessary diffusion Eddington factors are estimated by Monte Carlo, whereas [13] uses a pure Monte Carlo route. In both cases, the process is iterative. In the latter method, called MAGIC in the literature, multi group Monte Carlo could be used to speed up the iterations prior to a final, continuous energy iteration. MAGIC is implemented, for example, in OpenMC [14, 15] and SCONE [16].

In this work, the application of the random ray (RR) method to generate weight windows is tested with SCONE [16]. CADIS was implemented and tested on a detector problem, and both the route based on the forward flux and the FW-CADIS approach are adopted for GVR. The RR method, introduced in [17], is a stochastic version of the Method of Characteristics (MoC): it uses randomly sampled rays which vary between iterations rather than a fixed cyclic quadrature as is common in MoC. This implies that an initial angular flux per each ray cannot be stored; instead, each ray starts with an initial angular flux guess, but its contributions to the flux results are not tallied until it travels a fixed 'dead length'. After the dead

length, it is assumed that the error due to the original guess is sufficiently attenuated to accurately represent the local angular flux. Then, each ray is transported for a fixed 'active length', during which contributions to the scalar flux are recorded. The sum of the dead length and active length is also referred to as the 'termination length'. The use of a stochastic quadrature brings several benefits: the method is low memory, since angular fluxes at the boundaries are not stored; the model geometries are easily extendable to 3D, since there is no need for a closed quadrature. Additionally, the RR method does not discretise the angular variable, so it does not suffer from ray effects [18].

Although the use of randomly sampled rays introduces statistical uncertainties in the results, the error is distributed relatively uniformly in space and energy unlike in Monte Carlo [19]. This makes it ideal for supporting Monte Carlo VR calculations, and of great interest for applications such as radiation shielding and fusion neutronics. The RR method was only recently implemented in a fixed source calculation mode, and it proved to be computationally and memory efficient [20].

One of the drawbacks of common hybrid deterministic-Monte Carlo VR approaches is that a new geometry and input file have to be written to convert the Monte Carlo model to a deterministic model, unless the two codes are well integrated. However, due to the geometric flexibility allowed by on-the-fly ray tracing, RR can also directly use CSG Monte Carlo geometries, simplifying the VR process by removing (or lessening) the burden of constructing two simulation domains and mapping between them. The most significant difference between geometries is that RR requires a relatively fine geometric discretisation while Monte Carlo needs none, but this discretisation can be straightforwardly imposed by introducing a Cartesian grid onto the Monte Carlo geometry [21]. Additionally, model-dependent multi group cross sections must be generated for the RR geometry.

The rest of this paper will describe and evaluate the use of RR for Monte Carlo VR in more detail and provide numerical results for a detector problem and a deep penetration shielding problem. In the case of GVR, the two routes employing RR will be compared with MAGIC, also implemented in SCONE.

II. METHODOLOGY

In this section, the variance reduction methods tested and compared are described, together with the details of their implementation in SCONE.

II.A. CADIS

The CADIS methodology was introduced in [3], aiming to obtain zero variance results at a detector tally in a Monte Carlo calculation. CADIS prescribes which statistical weight Monte Carlo particles should have in order to obtain zero variance results at the detector of interest, as a function of the adjoint flux $\psi^\dagger(\mathbf{r}, E, \Omega)$, where $\mathbf{r}$ is a point in space, E is neutron energy, and Ω is the neutron flight direction. For this purpose, the adjoint source $q^\dagger(\mathbf{r}, E)$ is equivalent to the cross section at the detector of interest, $\Sigma_\mathrm{d}(\mathbf{r}, E)$. The optimal particle weight distribution, $w(\mathbf{r}, E, \Omega)$, is then the inverse of the adjoint flux normalised by the response function of interest, R,

$$w(\mathbf{r}, E, \Omega) = \frac{R}{\psi^\dagger(\mathbf{r}, E, \Omega)}, \quad R = \int_{\mathbf{r},E,\Omega} q^\dagger(\mathbf{r}, E)\psi(\mathbf{r}, E, \Omega) d\mathbf{r} dE d\Omega \; . \tag{1}$$

However, in practice, there are several limitations that prevent the method from being applied exactly. For instance, the adjoint flux is not known exactly, but it is often estimated with a deterministic solver (in this case, the RR solver in SCONE). This introduces an approximation with respect to continuous energy Monte Carlo. Furthermore, an additional approximation is often introduced where the adjoint angular flux is replaced with the adjoint scalar flux, $\phi^\dagger(\mathbf{r}, E)$. This is because the scalar flux is cheaper and easier to estimate (the global angular flux is rarely produced by transport codes due to its memory burden), but it neglects the importance associated with a neutron's direction. As a consequence, the weight definition in Eq. (1) reduces to

$$w(\mathbf{r}, E) = \frac{R}{\phi^\dagger(\mathbf{r}, E)}. \tag{2}$$

Then, the Monte Carlo source is generated from the biased distribution,

$$\hat{q}(\mathbf{r}, E) = \frac{\phi^\dagger(\mathbf{r}, E)}{R} q(\mathbf{r}, E) \; , \tag{3}$$

and particles are initialised with weights $w(\mathbf{r}, E)$.

Additionally the adjoint flux, and hence the weights, are not known as a continuous distribution, but they are discretised over space and energy and used in Monte Carlo as a weight window mesh. Because of these approximations, zero variance results are not realistically achieved. However, the CADIS method greatly helps in reducing the variance of Monte Carlo results at a detector compared to analog calculations; the reduction of the variance depends on parameters like the phase space mesh used for the weight windows, chosen ad-hoc for each model, and the quality of the deterministic adjoint calculation. The production of weight windows from CADIS-produced optimal weights is also not specified by the method – either the user or a code must decide on the 'width' of the weight windows. Similarly, the Russian Roulette survival weight is a free parameter, although it is often set as the local optimal weight. Finally, the settings of the Monte Carlo code can significantly affect efficiency, for example, the maximum number of particle splits allowed per particle or per family of particles.

II.B. FW-CADIS

CADIS is applicable to problems where the results of interest are at an individual detector. When the goal is to minimise the variance at multiple detector locations, or everywhere in the model (GVR), FW-CADIS is used instead. In this case, the adjoint source is weighted with the forward flux, $\phi(\mathbf{r}, E)$, at the locations of interest,

$$q^{\dagger}(\mathbf{r}, E) = \begin{cases} \frac{1}{\phi(\mathbf{r},E)} & \text{at the detectors} \\ 0 & \text{otherwise} \end{cases} . \tag{4}$$

This forward weighting is necessary to obtain an approximately uniform variance across all detectors. The forward calculation is usually performed by a deterministic solver, in this case RR. The rest of the method proceeds identically to CADIS.

II.C. Forward flux-based GVR

In the case of GVR, the weight window maps can be constructed directly from a forward flux estimate. The logic is the following, as explained by [12]: the distribution of 'physical' particles, $N(\mathbf{r}, E)$, is equal to the distribution of Monte Carlo

particles, $M(\mathbf{r}, E)$ multiplied by the statistical weight distribution:

$$N(\mathbf{r}, E) = M(\mathbf{r}, E)w(\mathbf{r}, E) \tag{5}$$

In the case of GVR, the goal of the calculation is to obtain a uniform uncertainty over the whole phase space. If the distribution of Monte Carlo particles is uniform in phase space ($M(\mathbf{r}, E) \simeq$ constant), the number of scores to the tallies will also be approximately constant. This can be achieved by imposing statistical weights that are proportional to the physical particle density. However, in neutron transport calculations it is customary to calculate the scalar flux rather than the particle density. These are two distinct physical quantities, but in practice the scalar flux is used:

$$w(\mathbf{r}, E) \propto N(\mathbf{r}, E) \simeq \phi(\mathbf{r}, E) \ . \tag{6}$$

As mentioned previously, this logic is applied in MAGIC [13]. MAGIC is an iterative, fully Monte Carlo method, where the scalar flux results from a Monte Carlo run are used to produce weight windows for the following run. The target weights are normalised by the maximum flux value in the system:

$$w(\mathbf{r}, E) = \frac{\phi(\mathbf{r}, E)}{\max(\phi(\mathbf{r}, E))} \ . \tag{7}$$

The need for more than two iterations (one to generate weight windows and one to produce final results) arises when particles are not able to populate the whole geometry in the first analog calculation. The termination criterion for the MAGIC iterations is whether scores for the forward flux are obtained in every tally region in the geometry. Afterwards, a final calculation, using the last weight windows generated, is performed to produce results. MAGIC is convenient because only one code has to be used to produce and use the weight windows; however, performing several Monte Carlo iterations might be computationally expensive, and the use of the forward rather than adjoint flux implies that the weight windows do not contain information about particle importance.

In this paper, the forward flux approach to generating weight windows was used with MAGIC and in combination with RR as well (labelled 'RR - flux' in Section III.A). In this case, a forward RR calculation is performed to produce the

forward scalar flux, then applied to Monte Carlo target weights as in Eq. (7). The main advantage of this method compared to MAGIC is that only one RR run is required to obtain results everywhere in the phase space, rather than several Monte Carlo iterations. On the other hand, this entails the practical disadvantages of discretisation and cross section generation.

II.D. Implementation in SCONE

The methods described so far were implemented in SCONE, the Monte Carlo code under development at the University of Cambridge [16]. The capability to read and apply weight windows was introduced in the past by [22]. By default, the weight windows target weight corresponds to the Russian roulette survival weight; the lower and upper weight windows are then a factor of 2 respectively lower and higher. A sensitivity study on the width of the weight windows was not performed, as these values are the default of MCNP [4] and as such are commonly used. When SCONE is run with OpenMP parallelism and in combination with weight windows, secondary particles (produced from splitting) are sent to a particle buffer common between multiple threads. The common buffer, together with a limit on the maximum number of splits per particle equal to 1000, is used to improve parallel performance and mitigate problems with long histories [23].

Implementing MAGIC simply consisted of performing multiple Monte Carlo fixed source iterations; at the end of an iteration, the scalar fluxes scored in the user-defined weight window mesh are processed and used as weight windows for the next iteration. When each tally bin has a non-zero score, the code runs the last iteration, as done in [13]. It is conceivable that the obtained optimal weight in a given window might have a large uncertainty. In this work, no threshold on the maximum uncertainty was introduced, nor were there any other treatments to mitigate the effects of these uncertainties. Once the particles have populated the whole phase-space, performing additional MAGIC iterations without increasing the particle population would only marginally improve the uncertainty of the calculated weight windows, if at all. While an alternative would be to perform another iteration with an increased particle population, the excessive cost of additional Monte Carlo calculations that do not contribute to the final results makes this undesirable. However, in order to allow the user the flexibility to choose how much time to spend

performing the MAGIC iterations and the desired precision in the generated weight windows, the possibility to vary population size between iterations was implemented in SCONE, as well as the option to run the convergence iterations with multi group cross sections and switch to continuous energy cross sections for the final iteration. By default, MAGIC is run with continuous energy cross sections and the particle population in the iterations prior to final results generation is half the population of the final run.

Since recently, SCONE includes a RR solver for eigenvalue [19] and fixed source calculations [20]. For more information about the theory of the random ray method for fixed source problems, the SCONE implementation and the performance compared to multi-group Monte Carlo, the reader can refer to [20].

The main modification required to implement CADIS was to convert a forward random ray solver into an adjoint one. This is simply done by swapping the fission production spectrum, χ, with the fission production cross section, $\bar{\nu}\Sigma_{\mathrm{f}}$, in the multi group fission production term:

$$\chi_g \sum_{g'} (\bar{\nu}\Sigma_{\mathrm{f}})_{g'} \phi_{g'} \rightarrow (\bar{\nu}\Sigma_{\mathrm{f}})_g \sum_{g'} \chi_{g'} \phi_{g'} \ .$$

Additionally, the scattering matrices have to be transposed to reverse the physics of up-scattering and down-scattering,

$$\Sigma_{\mathrm{s},l,g,g'} \rightarrow \Sigma_{\mathrm{s},l,g',g} \ ,$$

where l denotes the Legendre order of the scattering matrix corresponding to different orders of anisotropy. The most basic RR solver, which is used in this paper unless otherwise specified, models isotropic scattering ($l = 0$) and applies the flat source approximation (neutron sources are assumed to be spatially-flat across a mesh [24]).

In the case of FW-CADIS, a forward RR calculation is run first; SCONE uses the forward flux to generate an adjoint source, adjusts the cross sections as specified above to run an adjoint calculation, and returns the calculated adjoint flux.

II.E. NJOY cross section generation

The multi group cross sections used for the RR calculations (and multi group MAGIC in one instance, see Sec. III.A.3) were produced with the nuclear data processing code NJOY [25]. NJOY was chosen as a quick and reliable way to generate cross sections, under the assumption that Monte Carlo generated cross sections would not be available under realistic use-cases. The production route used the NJOY modules RECONR, BROADR, GROUPR, and CCCCR for formatting. The unresolved resonance reconstruction module was not called, since the unresolved resonance treatment was not used in the reference Monte Carlo calculations. The thermal cross section generation module for $S(\alpha, \beta)$ treatments was also ignored, despite one of the test problems being composed of water, because it complicated the CCCCR formatting excessively. This is a major approximation, that could impact the quality of the weight windows in the water problem.

The weighting spectrum adopted was a combination of a fast fission spectrum, a $1/E$ intermediate spectrum, and a Maxwellian thermal peak. Multi group cross sections were generated material-wise, since further spatial discretisation is impossible when the flux energy spectra are not known exactly. The background cross section of isotope i, σ_0^i, with atomic density N_i in a mixture composed of isotopes j, necessary to self-shield resonances, was based on the potential scattering cross sections σ_p:

$$\sigma_0^i = \frac{1}{N_i} \sum_{j \neq i} \sigma_p^j N_j \ . \tag{8}$$

The multi group cross sections were returned by NJOY in the ISOTXS format, and then processed into a SCONE input. The ISOTXS format is limited mainly because it does not allow for up-scattering in the scattering matrices and it did not return (n,2n) information. The computing time required to produce this data was negligible.

III. NUMERICAL RESULTS

In this section, numerical tests to assess the performance of RR-based VR methods were run for a detector problem, as well as a GVR problem. To compare

different VR schemes, a figure of merit (FoM) was defined:

$$\mathrm{FoM} = \frac{1}{t \cdot \overline{\sigma^2}} \,, \tag{9}$$

where t is the calculation time in minutes, and $\overline{\sigma^2}$ is the mean over the relative variances of the tallies of interest. All calculations were run on a Intel(R) Xeon(R) Platinum 8276 node on 56 cores with OpenMP parallelism.

III.A. Global Variance Reduction

The application of RR to GVR was tested on a deep penetration shielding problem. The model geometry and dimensions are shown in Fig. 1. There is an external 2 MeV neutron source, isotropic and uniformly distributed within a UO_2 square cylinder (in pink in the picture), where the uranium consists of only ^{238}U. The source is contained inside a concrete cask, placed in a room filled with air. The room is separated from an adjacent room by a 2 m thick concrete wall. The adjacent room contains a water cylinder representative of a person (dark blue in the figure). The isotopic compositions of the materials present in the model are reported in the Appendix. The whole model geometry spans 6.4 m × 2.4 m × 3.4 m. Vacuum boundary conditions are applied on all sides of the problem.

The weight window mesh used is Cartesian, with sides of 20 cm in all directions. This choice is consistent with values found in the literature [13]. Additionally, a sensitivity study performed varying the mesh cell sides size to 10 cm did not show large differences in the FoM, while using 5 cm deteriorated performance slightly. The energy group structure chosen for the RR calculations was ECCO33, normally used for fast reactor systems, and the overlaid geometrical discretisation (additional to that from the Monte Carlo geometry cells) was Cartesian, with sides of 5 cm in all directions (the total number of cells was ∼424,000). The RR parameters used in all calculations are: 50000 rays per cycle, 500 active cycles, 700 inactive cycles, 770 cm dead length, and 1770 cm termination length. These settings are conservative: the ray density is such that nearly all cells are hit each iteration and the dead length is larger than the corner-to-corner distance of the problem, ensuring a vacuum boundary condition hit and removal of error from the initial angular flux guess. The number of inactive cycles was obtained by trial-and-error, inspecting the

flux values in regions distant from the source to ensure they have stopped changing significantly, while the active cycles were chosen to ensure reasonably low statistical error in the reported fluxes. The automation of RR convergence metrics in fixed source problems is an area of ongoing research.

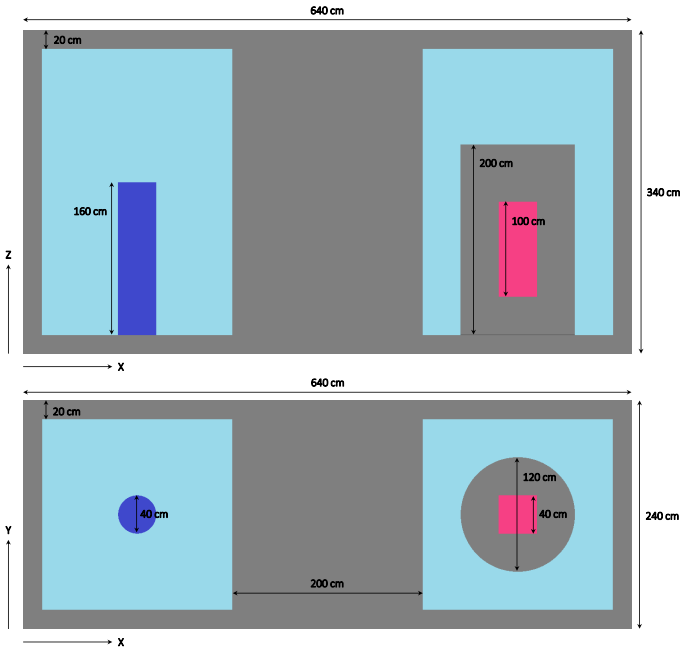


Fig. 1. Model geometry: light blue is air, dark blue is water, grey is concrete and pink is the UO_2 source

III.A.1. Population scaling

Different weight windows can greatly affect the run time of Monte Carlo simulations: the larger the weight attenuation throughout the geometry, the more splitting happens and the more particles have to be transported. In order to compare GVR methods that produced different weight windows, and to prevent excessively long computational times, the initial Monte Carlo particle populations of different runs were adjusted to keep the run time approximately constant. This led to using 3 billion particles in the analog calculations, and varying between ∼5 and ∼20 million particles for the variance reduction calculations.

Theoretically, the FoM introduced in Eq. (9) should be independent of the particle population used. However, when running numerical tests on the deep penetration problem presented in Fig. 1, this did not seem to be the case and had to be

verified. In order to ensure correct scaling of the FoM, several Monte Carlo instances with different particle populations were run with FW-CADIS generated weight windows. In order to isolate the Monte Carlo noise from the RR noise, the same weight windows were used in all cases. For each particle population, 10 different instances were run with different random number seeds. The results of this scaling test are shown in Fig. 2, where the error bars correspond to 1 standard deviation calculated between the 10 instances. Fig. 2 shows that the computational time scales exactly as expected with the particle population. However, while the calculated scaling of the mean error and FoM are within 3σ of the theoretical scaling, large variations are present between different instances of the same case. The large fluctuations in the mean standard deviation are driven by the high variance results on the left of the concrete shield, i.e., furthest from the source, where a large amount of particle splitting happens. As an example, the large range of run times and mean standard deviations of 10 Monte Carlo instances run with an initial population of 20 million particles is shown in Fig. 3.

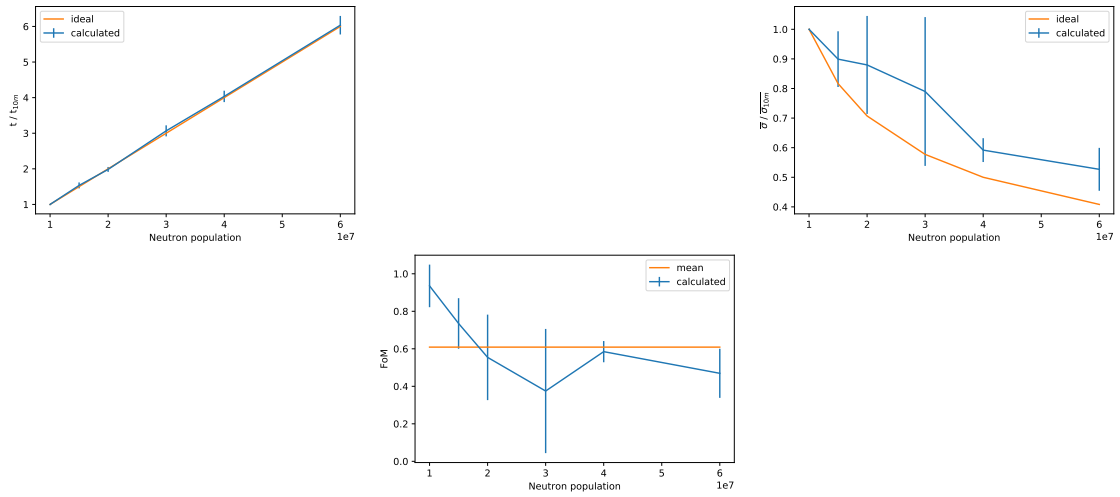


Fig. 2. Scaling of the average MC run time (left), mean standard deviation (centre) and FoM (right) with varying neutron population. The error bars correspond to 1σ standard deviation between 10 independent runs

While the correct population scaling seems to be preserved, in order to try and account for the high variance between individual runs, two measures where implemented: each case is run in 20 independent instances, each with independently generated weight windows, and the performance statistics are provided with a stan-

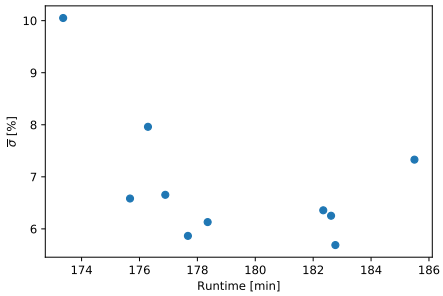


Fig. 3. Example of run time and mean standard deviation variation for the case of 20 million initial particles, with weight windows generated with FW-CADIS

dard deviation calculated from these repetitions.

III.A.2. Methods comparison

A sample of the mean flux and relative standard deviation profiles generated with FW-CADIS is shown in Fig. 4. The tally discretisation is a Cartesian grid with sides of 10 cm. The flux profile is normalised to have a maximum value of 1; on the left of the shield it is attenuated by approximately 14 orders of magnitude, which is a challenge for GVR and can lead to uncertainties in the water cylinder of up to ∼10% and ∼20% in the air, even when weight windows are being used.

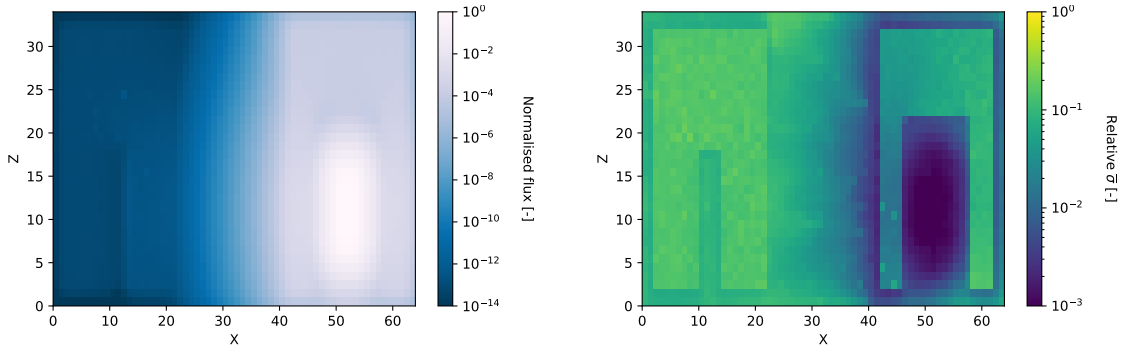


Fig. 4. Average flux and relative standard deviation in the central xz-slice of the geometry, from one of the instances simulated with FW-CADIS weight windows

The comparison between the performances of the different methods implemented, including an analog run, MAGIC, and RR generated weight windows (both forward and FW-CADIS), is presented in Table I. In all cases, applying GVR does not bias the flux results. In the table, the numbers presented are the mean between

the 20 independent instances, and the numbers after the ± correspond to 1σ standard deviation. Two variations of FoM are included: without and with the time taken to generate weight windows. The former more relevant if one generates weight windows only once and then performs several MC runs whereas the latter is more relevant if one is interested in an individual MC run.

In the analog runs, particles only reached about half of the tallied regions in the geometry; in the calculation of the FoM, the unreached cells were included with a standard deviation of 100%. In terms of FoM, the analog results are around 50 times worse than the methods employing GVR. However, the performance is quite consistent, i.e., the FoM has a low standard deviation. MAGIC was run entirely with continuous energy cross sections, and it took 4 iterations overall. The iterations needed to converge are run with half the particle population of the final iteration; still, the time taken to generate weight windows is significant, as can be seen from the drop in the FoM which takes into account the weight window generation time. The MAGIC results also have large fluctuations between independent runs, as can be seen by the standard deviation of the two FoMs which are ∼40%. As for the RR approach based on the forward scalar flux, the RR calculation itself is extremely fast. However, the mean realative variance and standard deviation are the highest of all cases, as is the standard deviation of the performance indicators between the 20 independent runs. FW-CADIS has the best FoMs and reasonable standard deviations of the performance indicators.

TABLE I
Performance comparison between different GVR cases. The numbers presented are the mean values between 20 independent runs, and those after the ± are the 1σ standard deviation

	Analog	MAGIC	RR - flux	FW-CADIS
WW gen. time [min]	-	127.9 ± 18.6	6.6 ± 0.06	13.2 ± 0.12
MC run time [min]	99.0 ± 2.9	108.4 ± 6.8	99.5 ± 16.3	102.4 ± 4.2
Relative $\bar{\sigma}$ [%]	55.7 ± 0.04	7.9 ± 1.1	10.7 ± 2.5	6.7 ± 0.4
Relative $\bar{\sigma^2}$ [%]	53.0 ± 0.06	1.2 ± 0.4	1.9 ± 1.1	0.9 ± 0.1
FoM	1.91e-02 ± 5.6e-04	0.80 ± 0.30	0.52 ± 0.31	1.08 ± 0.14
FoM - with WW time	-	0.37 ± 0.14	0.48 ± 0.29	0.96 ± 0.12

A more detailed visualisation of the distribution of the standard deviation of the flux tallied over the whole geometry in the different cases is shown in Fig. 5. As

mentioned earlier, the analog calculation has a large fraction of tallies that did not get any score, assigned with a 100% standard deviation. As for the other methods, FW-CADIS has the largest fractions of geometrical cells with a low stochastic error, while the RR approach based on the forward flux has the highest number of cells with uncertainties around and above 20%. The cells with uncertainties above 20% are mostly found in the air at the left of the shield.

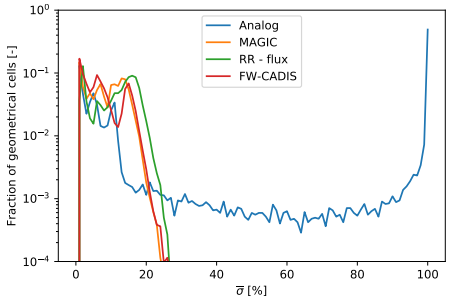
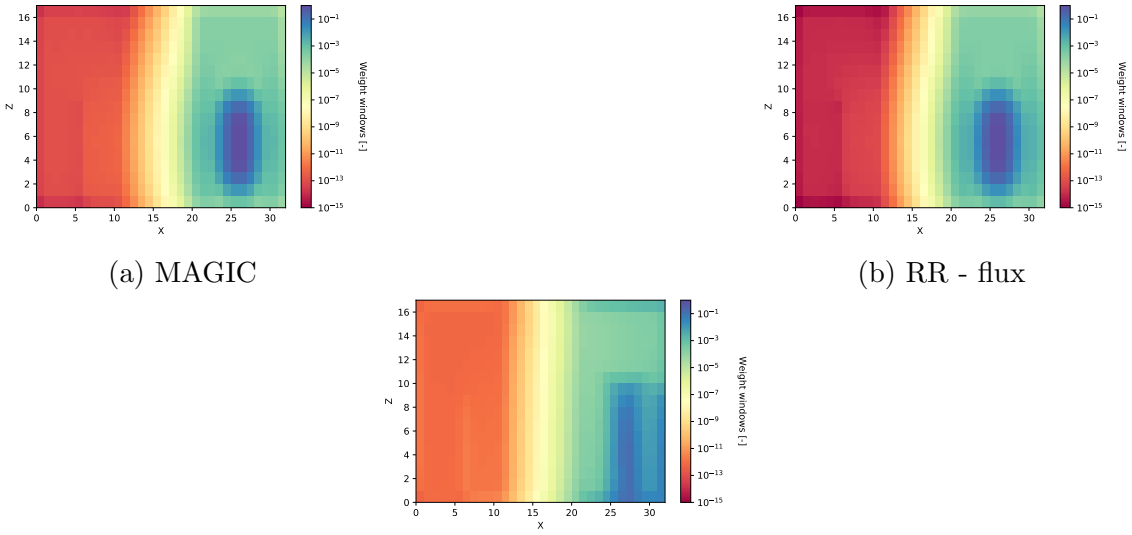


Fig. 5. Standard deviation distribution in the scalar flux tallied over a 10 cm × 10 cm × 10 cm mesh for the different cases

Fig. 6 reports a 2D slice of the weight windows produced by the GVR methods employed so far. The two routes based on the forward flux produce weight windows with a similar shape at the source and in the shield. However, the RR - flux (which in this case is directly proportional to the weight windows) is more heavily attenuated after the shield. The difference is due to the approximated physics captured in the NJOY multi group cross sections, and the discretisations imposed by RR. The strong attenuation throughout the shield might explain the higher variance observed in Fig. 5 for the the RR - flux case: the high variance might be a consequence of particles with different statistical weights scoring in the same tally bin. When, for example, fast energy particles go through the shield skipping over several weight windows cells at once, they collide and contribute to the local tally with a weight which could be several orders of magnitude higher than the local weight window values. This effect is accentuated the steeper the weight window gradient is. It is possible that this could be mitigated by adopting energy-dependent weight windows; however, this was not tested in this work.

FW-CADIS produces visibly different weight windows: in the vicinity of the source, the target weights are notably higher on the right compared to the forward flux methods. The use of the adjoint flux captures the fact that the region to the

right of the source is less important to the overall solution compared to the particles directed towards the shield. Additionally, the weight attenuation throughout the shield is the lowest between the three cases.



(a) MAGIC

(b) RR - flux

(c) FW-CADIS

Fig. 6. Weight windows in the central xz-slice of the geometry calculated by MAGIC (left), RR - flux (centre) and FW-CADIS (right)

The ratio between the maximum and minimum weight values is also shown in Table II, together with indicators of the stochastic noise in the weight windows. These indicators are the mean and maximum relative difference between weights calculated in symmetric regions of the geometry. These two regions are located on top of the room on the left side of the shield, in air. In particular, the comparison is between points: $2 < x < 11$, $y = 2$, $z = 16$ and their symmetric equivalents: $2 < x < 11$, $y = 11$, $z = 16$. Due to the y-symmetry of the geometry, the weights calculated at symmetric y coordinates should be the same in a specific calculation. Table II shows non-zero differences, which are entirely due to stochastic noise.

MAGIC weight windows are the noisiest among the three methods in the region considered. The RR solution has low stochastic noise, since the random rays are sampled uniformly throughout the geometry (unlike particles which originate from the source and are attenuated): as shown by [20], the standard deviation profile produced by RR is roughly uniform throughout the geometry. FW-CADIS is slightly noisier, since it consists of two RR runs, where the result of the first one is used as a

source for the second one. Clearly, the noise in the weight windows does not explain the trends observed in Table I, which showed that RR based on the forward flux had the worst performance. On the other hand, the weight attenuation decreases from RR, to MAGIC, to FW-CADIS. In this case, it would seem that the driver for a better or worse weight windows performance is the amount of splitting and low weight particles transported.

TABLE II
Ratio between maximum and minimum weight values, mean and maximum relative difference between weight values in symmetric cells (difference between $(2 < x < 11,\ y = 2,\ z = 16)$ and $(2 < x < 11,\ y = 11,\ z = 16)$). All the values reported were averaged over 20 independent instances

	MAGIC	RR - flux	FW-CADIS
Ratio max/min ww	4.2e-15	3.1e-16	9.7e-13
Mean relative difference [%]	8.7	0.6	4.7
Max relative difference [%]	23.2	1.5	11.3

III.A.3. MAGIC variations

In this section, different variations to the MAGIC route were tested to check if the MAGIC performance could be improved by reducing the time required to generate weight windows. This was attempted in two ways: reducing the number of particles run in the iterations prior the last by a factor of 10 rather than 2, and using multi group Monte Carlo in all iterations except for the last one. The cross sections used for multi group Monte Carlo are the NJOY generated ones, used previously for the RR runs. The results for this study are shown in Table III, where "ce MAGIC $\times 2$" is the reference case, presented also in the previous section. Both variations take a considerably shorter time to generate weight windows. In the case of the WW generation particle population being a tenth of the final calculation population, the mean relative variance and standard deviation produced are in the same ballpark as the reference case, with a slightly longer run time. The FoM that includes the weight window generation time improves on average compared to the reference, but the two values are largely within uncertainty of each other. The multi group MAGIC case with the same population scaling as the reference (denoted by "mg MAGIC $\times 2$") took 5 iterations instead of 4 to complete, and shows the worst performance. The

large relative standard deviation, also visualised in Fig. 7, and variance, degrade both FoMs. All cases present large fluctuations in performance between the 20 independent runs, with the highest variations produced by the continuous energy case with ×10 population scaling.

TABLE III
Performance comparison between different MAGIC variations. The numbers presented are the mean values between 20 independent runs, and those after the ± are the 1σ standard deviation

	ce MAGIC ×2	ce MAGIC ×10	mg MAGIC ×2
WW gen. time [min]	127.9 ± 18.6	30.0 ± 3.9	46.1 ± 29.8
MC run time [min]	108.4 ± 6.8	151.3 ± 14.4	99.3 ± 25.6
Relative $\bar{\sigma}$ [%]	7.9 ± 1.1	7.1 ± 1.8	12.8 ± 1.7
Relative $\bar{\sigma^2}$ [%]	1.2 ± 0.4	1.0 ± 0.8	2.5 ± 0.7
FoM	0.80 ± 0.30	0.64 ± 0.47	0.40 ± 0.15
FoM - with WW time	0.37 ± 0.14	0.53 ± 0.39	0.27 ± 0.11

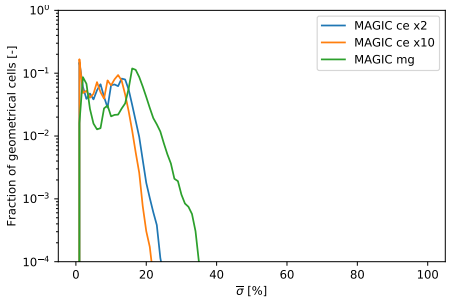


Fig. 7. Standard deviation distribution in the scalar flux tallied over a 10 cm × 10 cm × 10 cm mesh for the different cases for the different MAGIC variations

Table IV shows indicators of flux attenuation and noise in the weight windows for the three MAGIC variations. The flux attenuation is almost identical in the two continuous energy cases, while it is more pronounced in the multi group case. This was expected as multi group Monte Carlo employed the same multi group cross sections as RR, and the two methods are expected to produce similar solutions. As for RR in Section III.A.2, this might explain the poor performance of multi group MAGIC. In terms of stochastic noise, the worst case is by far the continuous energy with ×10 population scaling. It is conceivable that such high noise in the weight windows, which is an order of magnitude higher than in the other cases and

translates into very different weight windows between different independent runs, would cause a high standard deviation in the performance parameters too.

TABLE IV
Ratio between maximum and minimum weight values, mean and maximum relative difference between weight values in symmetric cells (difference between (x = (2,11), y = 2, z = 16) and (x = (2,11), y = 11, z = 16)). All the values reported were averaged over 20 independent instances

	ce MAGIC ×2	ce MAGIC ×10	mg MAGIC ×2
Ratio max/min	4.2e-15	4.1e-15	1.3e-16
Mean relative difference [%]	8.7	80.6	16.2
Max relative difference [%]	23.2	410	43.6

III.A.4. FW-CADIS variations

All the previous RR runs, including the FW-CADIS ones, were run with transport corrected P0 cross sections, and adopting the flat source (FS) approximation. However, as a result of recent work on the RR solver in SCONE, a higher order anisotropic scattering treatment was implemented, as well as the use of linear sources (LS) [26]. The aim of this section is to test whether higher fidelity RR solutions improve the performance of GVR calculations when using FW-CADIS. A summary of the results of this study, analysing the use of LS and anisotropy up to P3, is shown in Table V.

Comparing forward RR P0 scattering solutions with FS and LS against a multi group Monte Carlo solution from previous calculations, used as a reference, shows that using LS reduces the difference dramatically (from ∼90% average difference to ∼10% given the spatial discretisation used). However, the resulting weight windows do not improve the continuous energy Monte Carlo performance, which is almost identical. The higher relative standard deviation and variance, as well as the higher standard deviation between independent runs, are due to an outlier among the 20 Monte Carlo runs, which reported an average standard deviation of around 20%. Otherwise, the main difference is that the time required to generate weight windows with LS doubles given identical ray settings.

In terms of scattering anisotropy order, a comparison with a reference solution is impossible since multi group Monte Carlo cannot straightforwardly support

anisotropic scattering beyond P1. However, it was observed that the adjoint flux profiles correctly converge when increasing the anisotropy order: the average difference between the P1 and P3 adjoint fluxes is ∼23%, while this drops to ∼2% between P2 and P3. While the large difference between P1 and P3 indicates that this problem is highly anisotropic, higher fidelity FW-CADIS solutions do not help Monte Carlo performance. All the FoMs without WW generation time are largely within standard deviation of each other, and the large standard deviations between runs prevent drawing more definite conclusions. Furthermore, the significantly higher runtime of RR with higher-order anisotropy diminishes the FoM when including WW generation time, penalising the use of higher scattering fidelity.

TABLE V
Performance comparison between different FW-CADIS variations. The numbers presented are the mean values between 20 independent runs, and those after the ± are the 1σ standard deviation

	P0 - FS	P0 - LS	P1 - FS	P2 - FS	P3 - FS
WW gen. time [min]	13.1	28.4	24.3	49.2	88.7
MC run time [min]	102.4 ± 4.2	108.0 ± 5.1	135.8 ± 6.0	123.5 ± 3.2	127.8 ± 4.4
Relative $\bar{\sigma}$ [%]	6.7 ± 0.4	7.6 ± 1.8	6.0 ± 1.2	6.7 ± 1.0	6.3 ± 0.7
Relative $\bar{\sigma^2}$ [%]	0.9 ± 0.1	1.2 ± 0.6	0.7 ± 0.4	0.9 ± 0.3	0.8 ± 0.2
FoM	1.08 ± 0.14	0.75 ± 0.39	0.99 ± 0.48	0.91 ± 0.31	0.99 ± 0.27
FoM - with WW time	0.96 ± 0.12	0.59 ± 0.31	0.84 ± 0.41	0.65 ± 0.22	0.58 ± 0.16

III.A.5. Monte Carlo generated cross sections

The GVR runs employing RR based on the forward flux and FW-CADIS were repeated using Monte Carlo generated multi group cross sections instead of the NJOY ones. Although these are likely impractical to use in production runs, this allows us to investigate the impact of cross section fidelity on GVR performance. Monte Carlo cross sections are more accurate since they are generated including the correct flux spectra; additionally, this allows discretising the geometry into unique cross section regions for the same materials, which is not straightforwardly possible with NJOY given limited knowledge of the spatial variation in neutron spectrum a priori. For these calculations, the concrete shield is divided into 8 regions, and the source and concrete cask into two, as shown in Fig. 8. The multi group structure is ECCO33, as before. The SCONE multi group cross section generation route is

detailed in [22].

The expectation is that, while the NJOY cross sections are heavily approximated, the more representative Monte Carlo cross sections could help capture the importance distribution with higher accuracy, thus improving the performance of the weight windows. On the other hand the Monte Carlo cross sections, generated with a run adopting FW-CADIS weight windows, introduce stochastic noise. For instance, the uncertainty of the capture cross section in the thermal energy groups in relatively flat, and it increases from ∼0.05% to ∼5% from the rightmost layer of the shield to the leftmost. The mean uncertainties of the P0 scattering matrix in those regions are ∼6% and ∼15%, with minimum values of ∼0.05% and ∼6.7%.

The ratio between the weight windows produced with P0-FS FW-CADIS with NJOY cross sections and Monte Carlo cross sections is shown in Fig. 9, which highlights differences of up 2 orders of magnitude between the two solutions.

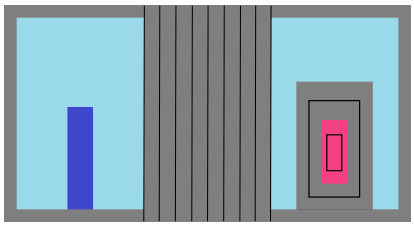


Fig. 8. Spatial discretisation applied for the generation of Monte Carlo multi group cross sections: the solid black lines separate adjacent homogenised regions

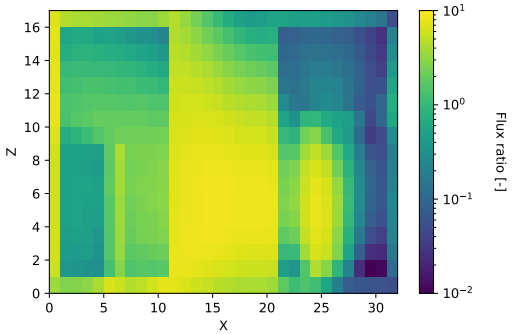


Fig. 9. Ratio between weight windows in the central xz-slice of the geometry calculated using FW-CADIS (FS with P0 scattering) with NJOY and Monte Carlo cross sections

The results are presented in Table VI. Compared against Table I, despite all

results being within one standard deviation of each other, the RR approach based on the forward flux has a higher average FoM and lower variations between independent runs. While the stochastic error in the weight windows is the same as before, since RR was run with identical parameters, the flux attenuation calculated decreased from 3.1e-16 to 8.7e-15, i.e., by almost a factor of 30. This could be the reason for the improved performance. While an accuracy comparison based on a reference continuous energy Monte Carlo solution is not straightforward, since the multiple approximations applied to RR might result in cancellation of errors, it must be noticed that the new attenuation factor is now closer to the one produced by MAGIC, shown in Table I. As for the FW-CADIS variations, all the performance parameters are within uncertainties of each other and of the results using NJOY cross sections. As a consequence, a clear improvement was not observed and, for this problem, there is little incentive to improve the fidelity of the cross section generation procedure.

TABLE VI
Performance comparison between RR and different FW-CADIS variations, using Monte Carlo generated cross sections. The numbers presented are the mean values between 20 independent runs, and those after the ± are the 1σ standard deviation

	RR - flux	P0 - FS	P0 - LS	P1 - FS	P2 - FS	P3 - FS
WW gen. time [min]	7.0	14.1	29.5	26.5	50.1	88.5
MC run time [min]	107.8 ± 3.1	108.3 ± 3.7	103.5 ± 3.3	131.3 ± 2.7	138.9 ± 3.7	129.9 ± 9.7
Relative $\bar{\sigma}$ [%]	7.0 ± 1.3	7.3 ± 1.3	7.0 ± 0.5	5.0 ± 1.0	5.9 ± 1.1	6.3 ± 0.5
Relative $\bar{\sigma^2}$ [%]	1.0 ± 0.4	1.3 ± 0.6	1.1 ± 0.2	0.6 ± 0.3	0.9 ± 0.5	0.9 ± 0.2
FoM	0.94 ± 0.38	0.72 ± 0.31	0.82 ± 0.13	1.33 ± 0.63	0.82 ± 0.51	0.82 ± 0.17
FoM - with WW time	0.88 ± 0.35	0.63 ± 0.27	0.64 ± 0.10	1.12 ± 0.53	0.60 ± 0.37	0.49 ± 0.10

III.A.6. Discussion of GVR results

To summarise, the performance of RR-based methods applied to GVR in the shielding model described is around 50 times better than analog Monte Carlo, performing similarly to MAGIC. When considering the FoM which takes into account the time required to produce weight windows, FW-CADIS is generally better. This is the case because MAGIC is penalised by the multiple costly Monte Carlo iterations needed to converge to a set of weight windows. Attempts to speed up weight window generation time, such as using multi group cross sections or drastically reducing the particle population used for weight window generation, failed to improve

the results. However, adopting MAGIC with a population scaling factor larger than 2 would likely produce better results if the particle population of the final run was increased, e.g., with the purpose of obtaining lower stochastic errors overall.

Although a comparison with FW-CADIS performed by a fully deterministic code, rather than RR, was not possible in this paper, the stochastic noise introduced in the weight windows by RR was minimal (as shown in Table II) and did not harm its performance. A more influential factor was how strong the attenuation of the flux, and hence the weights, was between the neutron source and the other end of the geometry. The stronger the attenuation, and the lower the weights, the more particles with extremely low weights are transported without significantly contributing to the final results. This factor, other than being model dependent, depends on which solver is used (RR with forward flux or FW-CADIS), which approximation is applied (FS vs LS, PN scattering) and which cross sections are being used (NJOY or MC generated), resulting in an uncontrollable and unpredictable performance. For instance, in this case, increasing the fidelity of the RR solver did not obviously improve the GVR performance.

More insightful comparisons were hindered by the large variations present between different repetitions of the same calculation. This high variability can be due to the independently generated weight windows being very different to each other because of stochastic noise, as it happened in the "ce MAGIC $\times 10$" case shown in Table IV, or by the large amount of local particle splitting present in deep penetration problems. This behaviour is consistent with the GVR methods review reported in [27]: the performance of GVR methods on deep penetration problems is highly model dependent and difficult to assess.

III.B. Variance Reduction to Detector

The performance of the CADIS algorithm implemented in the RR solver in SCONE was tested on a source-detector problem. The model, adopted from [28], is displayed in Fig. 10. In the figure, the 2 MeV neutron source is at the bottom left corner (in yellow in the picture), surrounded by fissile material (in pink). A detector (in orange) is placed on the other corner of the geometry, whose side is 100 cm long. In between is room temperature water. Also in this case, the multi group cross sections needed for CADIS are produced by two separate routes: with NJOY and

Monte Carlo. Once again, the Monte Carlo cross sections are more finely discretised than the NJOY ones: in the picture, the homogenised regions are highlighted by the different shades of blue. On the outside, framing the water, are steel (gray) and air (violet). The isotopic compositions of all the materials are reported in [28].

In this case, the result of interest is the neutron flux at the detector in two energy groups, thermal and fast, where the separation energy between the two is 0.0625 eV. The weight window mesh employed is Cartesian with sides of 1 cm, and is split into the same two energy groups as the detector.

The RR discretisation was Cartesian, with sides of 0.5 cm, resulting in 400,000 cells. The multi group cross section energy grid is CASMO12. The RR parameters used in all calculations are: 50000 rays per cycle, 1000 active cycles, 1000 inactive cycles, 150 cm dead length, and 500 cm termination length.

In comparison to the GVR problem, this was less computationally expensive. As a consequence, instead of varying the particle population between different cases to keep the Monte Carlo run time approximately constant, the particle population was fixed. The population used was 1.5 billion particles for each of the analog runs, and 1 billion particles for each variance reduction run. Also for this model, each case was repeated 20 times.

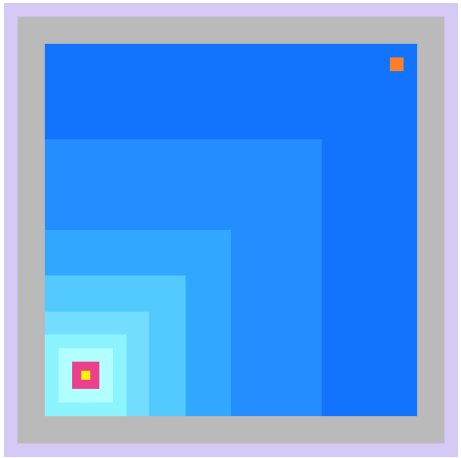


Fig. 10. Model geometry: the blue is water, orange is the detector, yellow the source, pink some fissile material, grey is steel and violet is air. The model is 2D, and its size is 100 cm $\times$ 100 cm. The different shades of blue highlight the discretisation used to generate multi group cross sections with Monte Carlo. The model is adopted from [28]

In this case, the performance comparison involves fewer methods, since an alternative way to produce weight windows (other than CADIS) was not available. As a consequence, only a comparison between analog Monte Carlo and CADIS was carried on, as well as a comparison between CADIS run with NJOY and Monte Carlo cross sections. The main differences between the two are the additional spatial discretisation following the Monte Carlo cross section generation regions, and the exclusion of thermal scattering data in NJOY. The results are reported in Table VII, where the RR solver used for CADIS adopted the default options, i.e., P0 isotropic scattering and FS approximation. The detector readings for the fast and thermal group are not reported, as they are all within uncertainties of each other. Additionally, their relative variance, used to calculate the FoMs, is not reported to keep the results compact. The RR run time, which was on the order of 2 minutes for all cases, was deemed negligible and not reported nor used to calculate the FoM. The mean relative standard deviation and weight windows in both energy groups for the case with NJOY cross sections are also shown in Fig. 11.

Table VII shows that the analog runs took the longest to run; this is partially justified by the $\times 1.5$ larger particle population, chosen to produce reference results, and by the fact that in the analog run thermal particles are modelled and transported everywhere in the geometry. This is not the case when variance reduction is employed, as shown in Fig. 11(a): the thermal particles are strongly directed towards the detector, while the very large weights on the other corners (Fig. 11(c)) ensure that particles that travel towards them are killed by Russian roulette. This is not the case in the fast group, where the weight windows are smoother (Fig. 11(d)) and particles are allowed almost everywhere (Fig. 11(c)), but the thermal group is more influential on run time because the thermal flux is around 3 orders of magnitude larger than the fast flux in this model.

The thermal and fast FoMs of CADIS are respectively 2 and 1 orders of magnitude larger than the analog ones. However, the standard deviations between independent cases is still extremely large, especially in the fast flux. The relative standard deviation at the detector in the thermal group is twice as low as that of the fast one, whereas this was not the case in the analog run: the use of weight windows preferentially reduced variance in the thermal group. On average, CADIS with NJOY cross sections performed better than with Monte Carlo cross sections. This is likely because the fluxes calculated with RR with Monte Carlo cross sections

were an order of magnitude more attenuated between the detector and the source. This results in more splitting of low weight particles, which can harm performance as seen in the previous section. The relative standard deviations in FoM between the two cross section sets are comparable.

TABLE VII
Performance comparison between different VR cases. The numbers presented are the mean values between 20 independent runs, and those after the ± are the 1σ standard deviation

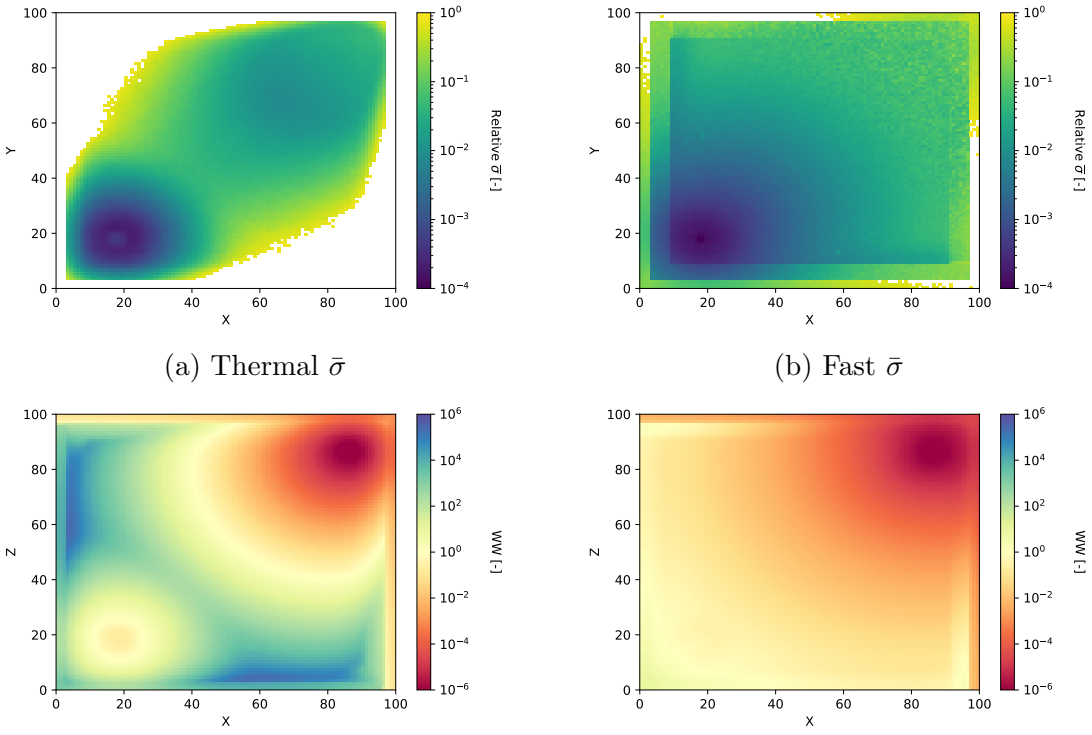
	Analog	CADIS (NJOY)	CADIS (MC)
MC run time [min]	303.3 ± 9.1	40.5 ± 1.0	129.1 ± 7.5
Relative $\bar{\sigma}$ thermal [%]	14.2 ± 2.1	2.9 ± 0.6	3.0 ± 0.6
Relative $\bar{\sigma}$ fast [%]	12.2 ± 1.3	6.0 ± 3.4	5.9 ± 4.5
FoM - thermal	0.16 ± 0.05	28.4 ± 11.5	8.4 ± 3.7
FoM - fast	0.22 ± 0.05	5.1 ± 7.6	1.4 ± 2.8

III.B.1. CADIS variations

Once again, an attempt was made to increase the fidelity of the RR solver to test whether more accurate CADIS weight windows improved performance. This study was done based on the Monte Carlo generated cross sections, because the lack of thermal scattering treatments was the accuracy bottleneck of the NJOY cross sections. The results of this study, including CADIS with P0 isotropic scattering and LS and with P3 scattering and the FS approximation, are shown in Table VIII. In this case, all results are extremely consistent, and no significant trend or difference was observed. The time needed to generate weight windows was ignored in the FoM calculation since is was between 2 and 4 minutes in all cases, and was considered negligible.

III.B.2. Discussion of VR results

The implementation and testing of CADIS in the SCONE RR solver for source-detector VR was successful, with FoM improvements by factors of ∼175 and ∼20 respectively in the thermal and fast energy groups. Surprisingly, using higher fidelity multi group cross sections in the RR solver did not improve VR performance, but deteriorated the FoMs by a factor of ∼3. Despite the large variations in performance



(a) Thermal $\bar{\sigma}$

(b) Fast $\bar{\sigma}$

(c) Thermal weight windows

(d) Fast weight windows

Fig. 11. Mean relative standard deviation of the flux (top) and weight windows (bottom) in the thermal (left) and fast (right) energy group, calculated using CADIS with isotropic scattering and FS approximation, using the NJOY cross sections

TABLE VIII
Performance comparison between different CADIS variations, performed with Monte Carlo generated cross sections. The numbers presented are the mean values between 20 independent runs, and those after the ± are the 1σ standard deviation

	CADIS P0 - FS	CADIS P0 - LS	CADIS P3 - FS
MC run time [min]	129.1 ± 7.5	135.2 ± 15.5	145.7 ± 14.1
Relative $\bar{\sigma}$ thermal [%]	3.0 ± 0.6	3.0 ± 0.9	2.7 ± 0.5
Relative $\bar{\sigma}$ fast [%]	5.9 ± 4.5	7.3 ± 5.5	6.3 ± 4.6
FoM - thermal	8.4 ± 3.7	7.7 ± 5.3	9.1 ± 3.1
FoM - fast	1.4 ± 2.8	0.88 ± 1.45	1.1 ± 2.2

between identical repetitions of the same calculation with different random number seeds, the difference in the mean FoMs is large enough to confidently state that high fidelity weight windows do not necessarily improve the results. This conclusion is reinforced by the fact that the NJOY generated cross sections lacked some very

important physics, i.e., the thermal scattering treatment in water, and that adding treatments like anisotropic scattering and LS did not make any difference.

IV. CONCLUSIONS

This paper investigates the application of the random ray method, the stochastic version of the deterministic Method of Characteristics, to Monte Carlo variance reduction. The random ray method was previously shown to have several characteristics that make it convenient for producing weight windows for Monte Carlo shielding problems: it produces a solution everywhere in the phase space with approximately uniform uncertainties, it is low memory and easily applicable to large 3D models, and does not suffer from ray effects. In this work, the CADIS and FW-CADIS methodologies were implemented in the random ray solver of SCONE, and tested on a GVR and a source-detector VR problem. Under the assumption that high fidelity multi group cross sections are not available in industrial applications, RR was run with computationally cheap NJOY cross sections. Then, the impact of this approximation on the performance of VR was assessed by comparing results produced with Monte Carlo generated cross sections. Additionally, the effects of higher fidelity RR solutions, modelling higher order scattering anisotropy rather than P0 transport corrected cross sections and LS rather than the FS approximation, were studied.

The GVR application was tested on a challenging deep penetration problem, including two meters of concrete shield. The RR based methods – FW-CADIS and weight windows generated from a normalised scalar flux calculated from RR – performed $\sim$50 times better than analog Monte Carlo, and comparably to MAGIC. The source-detector VR application was tested on a 2D problem mostly composed of water, with a neutron source and a detector in opposite corners of the geometry. The best CADIS results were $\sim$175 and $\sim$20 times better than analog Monte Carlo respectively in the thermal and fast energy group.

In both applications, improving the fidelity of the multi group cross sections used and of the RR solution did not improve the performance of VR. A more important factor was how low the particle weights become during a calculation: there is likely an optimum between ensuring that particles reach the desired tally location, while preventing excessive splitting and thus modelling of extremely low weight

particles. While these factors are model dependent and difficult to control, these results confirm that low fidelity cross sections are appropriate for producing weight windows. However, detailed comparisons were hindered by the high variability in performance of independent repetitions of the same runs.

Notably, the problems considered presently are 'bulk shielding' problems, i.e., there are no prominent streaming paths. Conversely, many shielding problems are dominated by particle streaming through small, low density regions. This is the case in 'maze' geometries, or in portions of fusion systems. In such streaming problems, obtaining accurate deterministic estimates of the flux (and so the optimal weights for variance reduction) is challenging. This is due to strong variations in the angular flux with direction and the resulting need for a refined angular quadrature. In this domain, one would expect the random ray method to be competitive with alternative deterministic solution methods due to the unique lack of angular discretisation. Methods such as MAGIC may also struggle in these problems as there is a much narrower range of directions (compared to the problems considered here) in which a neutron might travel that would result in contributing to a detector of interest. Future work may consider applying the random ray method to such streaming problems in anticipation of greater performance improvements.

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APPENDIX

The isotopic densities used for the materials in the deep penetration shielding model presented in Section III.A are reported below:

Material	Isotope	Density $[\frac{\text{atoms}}{\text{barn}\cdot\text{cm}}]$
Concrete	^{1}H	1.05015e-02
	^{10}B	3.59530e-04
	^{11}B	1.44651e-03
	^{16}O	3.96117e-02
	^{19}F	2.27070e-04
	^{23}Na	9.86759e-04
	^{24}Mg	1.77392e-04
	^{27}Al	4.44590e-04
	^{28}Si	2.20867e-03
	^{32}S	5.34762e-03
	^{39}K	4.79409e-05
	^{40}Ca	2.92693e-03
	^{55}Mn	6.82254e-06
	^{56}Fe	7.34675e-04
	^{64}Zn	1.89044e-04
	^{138}Ba	5.47725e-03
Source	^{238}U	0.02
	^{16}O	0.04
Water	^{1}H	0.0666
	^{16}O	0.0334
Air	^{14}N	4.14861e-05
	^{16}O	1.11693e-05
	^{40}Ar	4.78686e-07